3. Develop an Android application demonstrating complete Activity Lifecycle methods and analyze execution using Logcat.

AIM: To develop an Android application that demonstrates all the Activity Lifecycle methods and analyze their execution using Logcat**.**

THEORY

An **Activity** is a single screen in an Android application. Android manages the lifecycle of an activity through a series of callback methods. These methods are automatically invoked depending on the activity's state, such as launching, pausing, stopping, restarting, or destroying.

Android Activity Lifecycle Methods

1. **onCreate()**
   - Called when the activity is created.
   - Used for initialization and loading the UI.
2. **onStart()**
   - Called when the activity becomes visible to the user.
3. **onResume()**
   - Called when the activity comes to the foreground and is ready for user interaction.
4. **onPause()**
   - Called when another activity partially covers the current activity.
5. **onStop()**
   - Called when the activity is no longer visible.
6. **onRestart()**
   - Called after the activity has been stopped and before it starts again.
7. **onDestroy()**
   - Called before the activity is destroyed.

MainActivity.java

```
package com.example.lifecycleapp;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.util.Log;
public class MainActivity extends AppCompatActivity {

   String TAG = "ActivityLifecycle";
   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);
      Log.d(TAG, "onCreate() called");
   }

   @Override
   protected void onStart() {
      super.onStart();
      Log.d(TAG, "onStart() called");
   }

   @Override
   protected void onResume() {
      super.onResume();
      Log.d(TAG, "onResume() called");
   }
```

```
    @Override
    protected void onPause() {
        super.onPause();
        Log.d(TAG, "onPause() called");
    }

    @Override
    protected void onStop() {
        super.onStop();
        Log.d(TAG, "onStop() called");
    }

    @Override
    protected void onRestart() {
        super.onRestart();
        Log.d(TAG, "onRestart() called");
    }

    @Override
    protected void onDestroy() {
        super.onDestroy();
        Log.d(TAG, "onDestroy() called");
    }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">
   <TextView
      android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Android Activity Lifecycle Demo"
        android:textSize="22sp"
        android:textStyle="bold"/>
</LinearLayout>
```

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    package="com.example.lifecycleapp">
    <application
        android:allowBackup="true"
        android:label="LifecycleApp"
        android:supportsRtl="true"
        android:theme="@style/Theme.AppCompat.Light.DarkActionBar">
        <activity android:name=".MainActivity">
            <intent-filter>
                <action android:name="android.intent.action.MAIN"/>
                <category android:name="android.intent.category.LAUNCHER"/>
```

```
        </intent-filter>
    </activity>
</application>
</manifest>
```

Output

